Advancing Materials Development with Quantum AI and VQE Simulations of Majorana Edge States and Topological Phases

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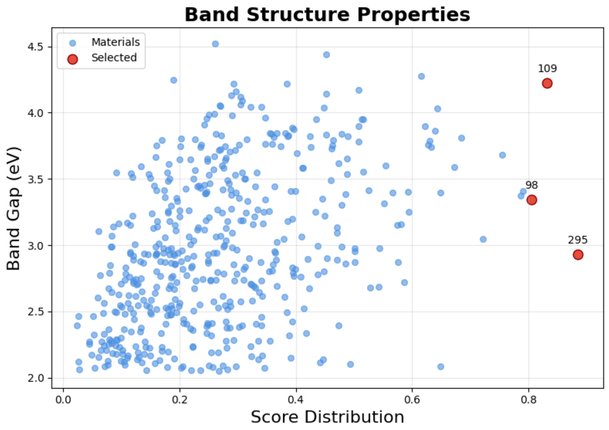
Introduction

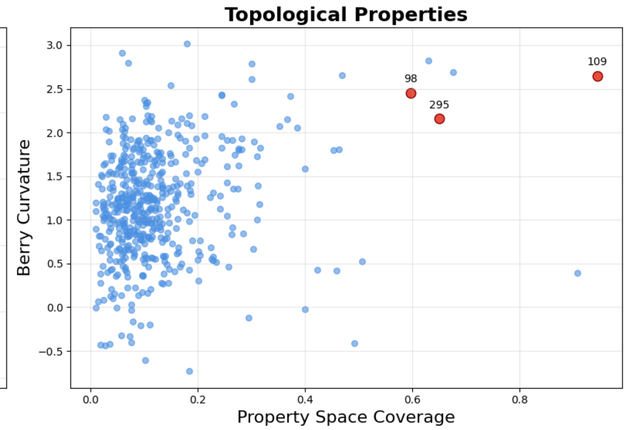
Superconducting quantum computing is costly due to its need for extremely low temperatures. To reduce these expenses, it is crucial to develop materials with topological properties that operate above 1K. The generation of synthetic materials combined with Variational Quantum Eigensolver (VQE) allows us to calculate Majorana edge states and topological phases, enabling identification of novel, suitable materials for topological quantum computing (QC).

Methodology

- Generate synthetic materials with various properties
- Calculate the band gap, berry curvature, coherence time, and gate fidelity
- Run the VQE on the top 3 materials by running through different values of μ, t, and Δ
- Run the classical optimization and compare with VQE results
- Cross-reference the other factors, environmental stability and manufacturing, and choose the best material for the real word

Material Analysis



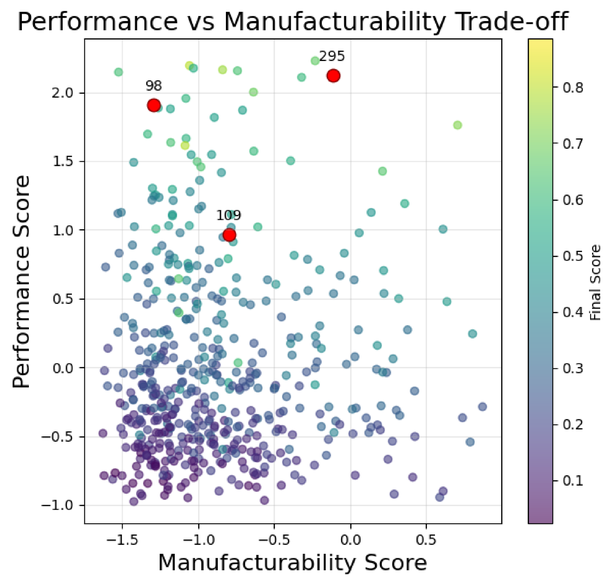


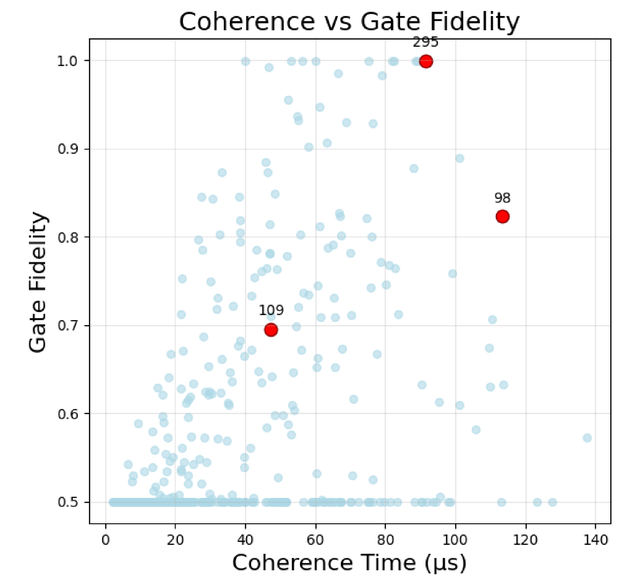
The overall score identifies the viability of each material to have edge states.

Kitaev-Chain Hamiltonian for N qubits:

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2}\right) + \sum_{j=1}^{N-1} \left[-t\left(c_{j+1}^\dagger c_j + c_j^\dagger c_{j+1}\right)\right] + |\Delta|(c_{j+1}^\dagger c_j^\dagger + c_j c_{j+1}),$$

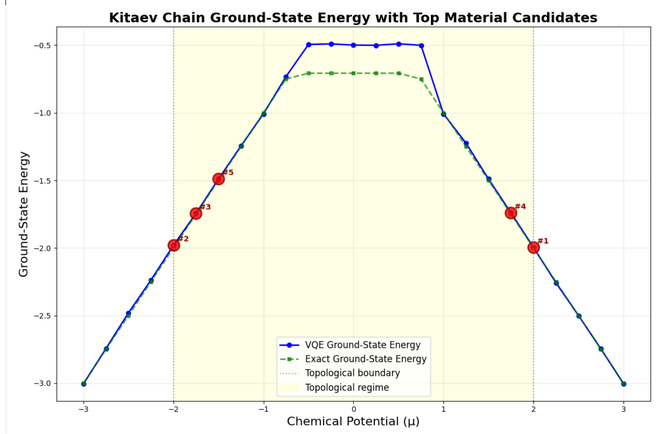
μ = chemical potential, t = energy for lattice hopping, and Δ = superconducting pairing





This trade-off and coherence time vs. gate fidelity shows the feasibility of manufacturing, as well as the quality of each material for topological QC.

Results



Material 295, Candidate 1

Band Gap: $2.93\ eV$

Energy gap: $1.2929\ eV$

VQE ground-state energy: $-1.9955\ eV$

Exact ground-state energy: $-2.0000\ eV$

Error: $0.004517\ eV$

Coherence time: $91.5\ \mu s$

Berry Curvature: 2.156

Applications

- Novel material design
- Environmental sustainability
- Enhanced product performance

Conclusions & Future Work

- VQE can handle more complex quantum systems more efficiently
- Density Functional Theory (DFT) to generate real, possible materials
- Simulate manufacturing process and implementation

Acknowledgements

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